MIL-C-36643A
12 July 1971
SUPERSEDING
MIL-C-36643
3 March 1970

MILITARY SPECIFICATION

CANNULA, TRACHEOTOMY, MODIFIED, JACKSON

This specification is mandatory for use by all the Agencies and Departments of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers tracheotomy tubes of the modified Jackson design, with and without endotracheal adapter, suitable for insertion into the trachea.

* 1.2 Classification. The tracheotomy tubes shall be of the following sizes and types as specified (see 6.2).

Size (Nominal Outside Diameter of Outer Tube in Inches, in the transverse axis)

2 - 0.236
3 - 0.274
4 - 0.314
5 - 0.351
6 - 0.393
7 - 0.440
8 - 0.474

Type I - Without Endotracheal Tube Adapter.

Type II - With Endotracheal Tube Adapter.

FSC 6515

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical; General Specifications.
MMM-A-250	Adhesive, Water Resistant (for Sealing Fiberboard Boxes).
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Set-Up.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing).

STANDARDS

Federal

FED-STD-151	Metal, Test Methods.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center (DPSC)

24083	Cannula, Tracheotomy, Modified; Jackson.

(Copies of specifications, standards and drawing, required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (A.N.S.I.)

A.N.S.I. Z79.2 Endotracheal-Tube Connectors and Adapters

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM F86 Recommended Practice for SURFACE REQUIREMENTS OF METALLIC SURGICAL IMPLANTS

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The Pharmacopeia of the United States of American (U.S.P.)

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa. 18042.)

3. REQUIREMENTS

* 3.1 Material. The tracheotomy cannula assembly (except pilot and bearing washer) shall be fabricated entirely from austenitic corrosion-resisting steel as specified below:

* 3.1.1 Inner and outer tubes, neck plate, swivel lock plate, inner tube anchor plate, swivel lock and endotracheal tube adapter. Shall be fabricated from type 304, class 3 material in accordance with Federal Specification GG-I-526. As an alternate, inner and outer tubes shall be fabricated from type 316 or 317, class 3 material in accordance with GG-I-526. Finished tubing shall have a final Rockwell hardness of B85 to B100 when tested as specified in 4.4.2.

3.1.2 Pilot (obturator). Pilots shall be fabricated from type 260, 270 or 360, class 8 material, half hard temper, in accordance with GG-I-526. Pilots shall be nickel plated or chromium plated over nickel.

* 3.1.3 Bearing washer. Bearing washer shall be fabricated from safe (nontoxic), durable, nonmetallic material possessing good wear characteristics, suitable for the purpose intended. The bearing washer shall be capable of withstanding repeated steam sterilization.

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Table II. Sampling Criteria for Tests (cont'd.)

Component	Characteristic	Req't	Test Procedure	Insp. level	AQL
Cannula Assembly	Boil Test	3.7	4.4.1.1	S-2	0.65
	Copper Sulfate Test	3.7	4.4.1.2	S-2	0.65
	Saline Test	3.7	4.4.1.3	S-2	0.65
	Sterilization Test	3.7.1	4.4.4	S-1	4.0
Pilot	Salt-Spray Test	3.7	4.4.1.4	S-2	0.65

* No defects permitted.
** A minimum of two samples shall be tested.

4.2.3 Raw materials. A sufficient quantity of each lot of each metal used in the construction of the tracheotomy cannulae shall be selected for chemical analysis.

4.2.4 Sample unit usage. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

* 4.3 Examination. The tracheotomy cannulae shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in Tables I and III. Inspection, however, shall not be restricted to defects listed in Table III. Any cannula in the sample containing one or more defects shall be rejected, and if the number of defective items in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

Table III. Classification of Defects

Categories	Defects
Critical	None defined.
Major	
101	Cannula not size indicated.
102	Cannula assembly not furnished complete with all specified components.
103	Inner tube extends beyond outer tube when locked into position.
104	Mating components fail to fit properly or seat properly when inserted into one another.
105	Action of swivel lock too stiff, too loose, or fails to lock inner tube securely to outer tube.
106	Components of cannula assembly not free from sharp edges or burrs (particularly distal ends of inner and outer tubes).
107	Components of cannula assembly not free from cracks, crevices, rough areas and tool marks.
108	Finish not bright, smooth and uniform, except for alternate finish specified in 3.5.

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Table III. Classification of Defects (cont'd.)

Categories	Defects
Major	
109	Welded or soldered joints not secure and free for fissures, flux, discoloration and excess solder or welding material.
110	Neck plate not free to move about outer tube.
111	Cannula assembly not permanently identified as specified in 3.9.
112	Cannulae assemblies not entirely fabricated from stainless steel (except for obturator and bearing washer).
113	Endotracheal tube adapters not permanently secured to proximal end of outer tube (type II items).
114	Bearing washer not part of assembly or fabricated from metal.
115	Swivel lock member does not possess two locking arms.
116	Inner and outer tubes not free from foreign matter.
117	Labels or printed markings on unit container not correct.
Minor	
201	Labels or printed marking on unit container not legible and free from defacing marks, scratches and tears.
202	Pilot not plated.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Corrosion tests.

4.4.1.1 Boil test. Tracheotomy cannulae shall be subjected to the boil test as described and specified in GG-I-526.

4.4.1.2 Copper Sulfate test. Tracheotomy cannulae shall be subjected to the copper sulfate test as described and specified in GG-I-526.

4.4.1.3 Saline test. Tracheotomy cannulae shall be scrubbed with soap and warm water, rinsed in hot water, followed by immersion into ethyl alcohol (95 percent) and dried. Cannulae shall then be completely immersed into Normal Saline Solution (U.S.P.) and while immersed, placed into an oven at a temperature of 37 ± 1 degrees Centigrade for a period of 7 days, removed from the oven and examined for corrosion while still immersed.

4.4.1.4 Salt-Spray test. Pilots shall be subjected to the salt-spray test as described and specified in GG-I-526.

* 4.4.2 Hardness test. Rockwell hardness of the tubes shall be determined in accordance with Method 244 of Federal Test Method Standard No. 151.

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* 4.4.3 Interchangeability test. Inner and outer tubes shall be completely separated, including the pilots. The inner and outer tubes shall be randomly reassembled. The inner tube shall be easily, freely and smoothly, capable of insertion into, and removal from, the outer tube without exertion of force and evidence of sticking or improper seating when the inner tube is secured to the outer tube by means of the swivel lock arrangement. The swivel lock shall readily secure the inner tube into proper position without the inner tube extending beyond the distal end of the outer tube when the assembly is held in the position as shown on the Drawing.

Pilots shall be randomly inserted into outer tubes. Pilots shall properly fit into the outer tubes by holding the retaining device at the proximal end of the pilots with the thumb. A portion of the olive shaped body shall extend beyond the distal edge of the outer tube, without falling completely through the outer tube opening.

* 4.4.4 Sterilization test. Sample units shall be placed into a suitable sterilizer (container) and steam sterilized at a minimum temperature of 270 degrees Fahrenheit for a period of 10 minutes. The tubes and the pilots shall then be removed from the sterilizer and allowed to cool to room temperature. This test shall be performed a total of 10 times on the same sample units.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. One cannula of the type and size specified constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	6 units	288 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

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5.1.3 Level A.

* 5.1.3.1 Unit package. Each unit shall be thoroughly cleaned and packaged in a bag fabricated from polyethylene seamless tubing with no wall thickness at any point below 0.00225 inches. Excess air shall be expelled or evacuated from the bag and closure effected by heat sealing. Each unit shall be further packaged in a box of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676 except that commercial colors are acceptable. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with 2 inch wide tape conforming to PPP-T-76. Closing of bottom flaps with adhesive conforming to MMM-A-250 or metal stitches is permitted.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.

* 5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

* 5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure of wood boxes shall be as specified in the appendix of the applicable box specification. Closure of fiberboard boxes shall conform to method II of PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

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5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The tracheotomy cannulae described herein shall be used to provide an airway through the trachea. All inner tubes and outer trachea tubes shall be interchangeable with each other within a given size and between types as designated herein.

6.2 Ordering data. Procurement documents should specify:

(a) Number, title and date of this specification.
(b) Size and type (see 1.2).
(c) Applicable Federal stock number.
(d) Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.3 This specification covers the following items listed in the Federal Supply Catalog:

Federal Stock Number	Item Identification
6515-518-5461	CANNULA, TRACHEOTOMY, Modified, Jackson, Size 2.
6515-518-5463	CANNULA, TRACHEOTOMY, Modified, Jackson, Size 2, with 15 mm Adapter.
6515-518-5464	CANNULA, TRACHEOTOMY, Modified, Jackson, Size 3.
6515-518-5467	CANNULA, TRACHEOTOMY, Modified, Jackson, Size 3, with 15 mm Adapter.
6515-518-5469	CANNULA, TRACHEOTOMY, Modified, Jackson, Size 4.

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6.3 (Cont'd.)

Federal Stock Number	Item Identification
6515-518-5472	CANNULA, TRACHEOTOMY, Modified, Jackson, Size 4, with 15 mm Adapter.
6515-518-5473	CANNULA, TRACHEOTOMY, Modified. Jackson, Size 5.
6515-518-5474	CANNULA, TRACHEOTOMY, Modified, Jackson, Size 5, with 15 mm Adapter.
6515-518-5475	CANNULA, TRACHEOTOMY, Modified. Jackson, Size 6.
6515-518-5477	CANNUAL, TRACHEOTOMY, Modified, Jackson, Size 6, with 15 mm Adapter.
6515-518-5478	CANNULA, TRACHEOTOMY, Modified, Jackson, Size 7.
6515-518-5481	CANNULA, TRACHEOTOMY, Modified, Jackson, Size 7, with 15 mm Adapter.
6515-518-5494	CANNULA, TRACHEOTOMY, Modified, Jackson, Size 8.
6515-518-5498	CANNULA, TRACHEOTOMY, Modified, Jackson, Size 8, with 15 mm Adapter.

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* 6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project Number 6515-1599

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the *Department of Defense*. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-C-36643A

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

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